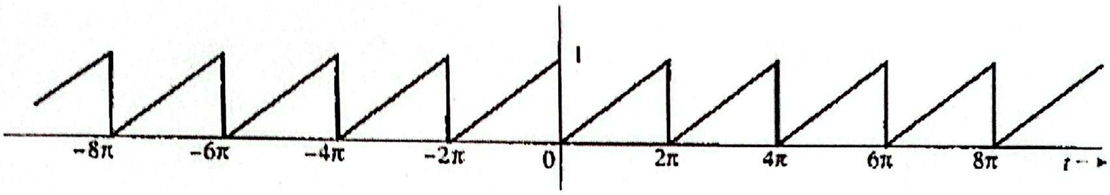
Habib University
Course Code: EE252
Course Title: Signals and Systems – Spring 2025
Instructor's Name: Tariq Mumtaz
Examination: Quiz 2 Date: 13th March, 2025
Total Marks: 50 (5% Course credit) Duration:60 minutes

CLO: 02; CDL: 4, PLO: 2

Q1[20]. Break down the given continuous-time periodic signal into its trigonometric Fourier series components and demonstrate that it can be expressed in terms of its components as follows:

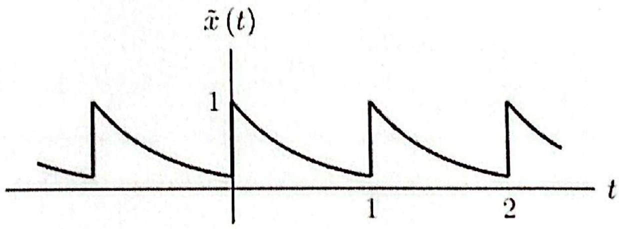
$$x(t) = \frac{1}{2} - \frac{1}{\pi}\left(sint + \frac{1}{2}\, sin2t + \frac{1}{3}\, sin3t + \cdots \ldots\right)$$



Q2[15] Shows that the EFS coefficient of a given signal is equal to: $\boldsymbol{Ck} = \frac{0.864}{2+j2\pi k}$

Also, sketch the amplitude and angle spectrum of C_k. Hint, Calculate C_k only at k = 0, infinity, and then interpolate the intermediate values.

$$\tilde{x}(t) = e^{-2t} \text{ for } 0 < t < 1 \text{ s.}$$



Q3[15]] Shows that the EFS coefficient of a given half-wave sinusoidal signal x(t) is equal to:

$$c_k = \frac{1}{4\pi(k-1)}\left[e^{-j\pi(k-1)} - 1\right] - \frac{1}{4\pi(k+1)}\left[e^{-j\pi(k+1)} - 1\right]$$

$$\tilde{x}(t) = \begin{cases} \sin(\omega_0 t)\,, & 0 \le t < T_0/2 \\ 0\,, & T_0/2 \le t < T_0 \end{cases}, \quad \text{and} \quad \tilde{x}(t+T_0) = \tilde{x}(t)$$

Formulae:

$$e^{ix} = \cos(x) + i\sin(x)$$

$$\int x\cos(ax)\,dx = \frac{1}{a^2}\left[\cos(ax) + ax\sin(ax)\right]$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\int x\sin(ax)\,dx = \frac{1}{a^2}\left[\sin(ax) - ax\cos(ax)\right]$$

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

Trigonometric Fourier series (TFS)

Synthesis equation:

$$\tilde{x}(t) = a_0 + \sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) + \sum_{k=1}^{\infty} b_k \sin(k\omega_0 t)$$

Analysis equations:

$$a_k = \frac{2}{T_0}\int_{t_0}^{t_0+T_0} \tilde{x}(t)\cos(k\omega_0 t)\,dt\,, \quad \text{for} \quad k = 1,\ldots,\infty$$

$$b_k = \frac{2}{T_0}\int_{t_0}^{t_0+T_0} \tilde{x}(t)\sin(k\omega_0 t)\,dt\,, \quad \text{for} \quad k = 1,\ldots,\infty$$

$$a_0 = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} \tilde{x}(t)\,dt \qquad \text{(dc component)}$$

Exponential Fourier series (EFS)

Synthesis equation:

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} c_k\, e^{jk\omega_0 t}$$

Analysis equation:

$$c_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} \tilde{x}(t)\, e^{-jk\omega_0 t}\,dt\,, \quad k = -\infty,\ldots,\infty$$

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46/50.

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Q.1) $(0,0)$, $(2\pi, 1)$

$m = \frac{1-0}{2\pi-0} = \frac{1}{2\pi}$ $\qquad \omega_0 = \frac{2\pi}{T_0} = \frac{2\pi}{2\pi} = 1$

$\boxed{x(t) = \frac{t}{2\pi}}$

$$a_0 = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\,dt$$

$$a_0 = \frac{1}{2\pi}\int_0^{2\pi} \frac{t}{2\pi}\,dt$$

$$a_0 = \frac{1}{2\pi}\left[\frac{t^2}{4\pi}\right]_0^{2\pi} = \frac{1}{2\pi}\left[\frac{4\pi^2}{4\pi}\right] = \frac{1}{2}$$

$$a_k = \frac{2}{2\pi}\left[\int_0^{2\pi} \frac{t}{\pi} \cos(k\omega_0 t)\,dt\right]$$

$$= \frac{1}{2\pi^2}\left[\int_0^{2\pi} t\cos(k\omega_0 t)\,dt\right]$$

$$= \frac{1}{2\pi^2}\left[\frac{1}{(k\omega_0)^2}\left[\cos(kt) + kt\sin(kt)\right]\right]_0^{2\pi}$$

$$= \frac{1}{2\pi^2 k^2}\left[\cos(2\pi k) + 2\pi k \underbrace{\sin(2\pi k)}_{0} - 1\right]$$

$$= \frac{1}{2\pi^2 k^2}\left[\underbrace{\cos(2\pi k)}_{1} - 1\right]$$

20

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$$b_k = \frac{2}{2\pi}\left[\int_0^{2\pi} t \sin(kt)\,.dt\right]$$

$$= \frac{1}{2\pi^2}\left[\frac{1}{k^2}\left[\sin(kt) - kt\cos(kt)\right]\right]_0^{2\pi}$$

$$= \frac{1}{2\pi^2 k^2}\left[\cancel{\sin(2\pi k)} - 2\pi k\cos(2\pi k)\right]$$

$$= -\frac{\cos(2\pi k)}{\pi k}$$

limit application

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$\boxed{T_0 = 1}$

Q.2) $c_k = \frac{1}{1}\int_0^1 e^{-2t} \cdot e^{-jk\omega_0 t} \, dt$

$= \int_0^1 e^{-2t - jk\omega_0 t} \, dt$

$= \left[\frac{e^{-2t - jk\omega_0 t}}{-2 - jk\omega_0}\right]_0^1$

$c_k = \frac{e^{-2 - jk\omega_0}}{-2 - jk\omega_0} - \frac{1}{-2 - jk\omega_0}$

$\boxed{\text{since } T_0 = 1}$ $\boxed{\omega_0 = 2\pi}$

$c_k = \frac{e^{-2 - j2\pi k}}{-2 - j2\pi k} - \frac{1}{-2 - j2\pi k}$

$c_k = \cancel{\frac{e^{-2}}{e^{j2\pi k}(-2 - j2\pi k)} - \frac{1}{-2 - j2\pi k}\left(\frac{e^{j2\pi k}}{e^{j2\pi k}}\right)}$

$c_k = \frac{-1}{2 + j2\pi k}\left(e^{-2 - j2\pi k} - 1\right)$

$= \frac{1 - e^{-2 - j2\pi k}}{2 + j2\pi k}$

$c_k = \frac{1 - e^{-2(1 + j\pi k)}}{2 + j2\pi k}$ bcs iti fist harmonic $k = 1$

$c_k = \frac{1 - e^{-2}\left(\overset{=1 \text{ for all } k}{\cos(2\pi)} - j\,\overset{=0 \text{ for all } k}{\cancel{\sin(2\pi)}}\right)}{2 + j2\pi k}$

$\boxed{c_k = \frac{0.865}{2 + j2\pi k}}$

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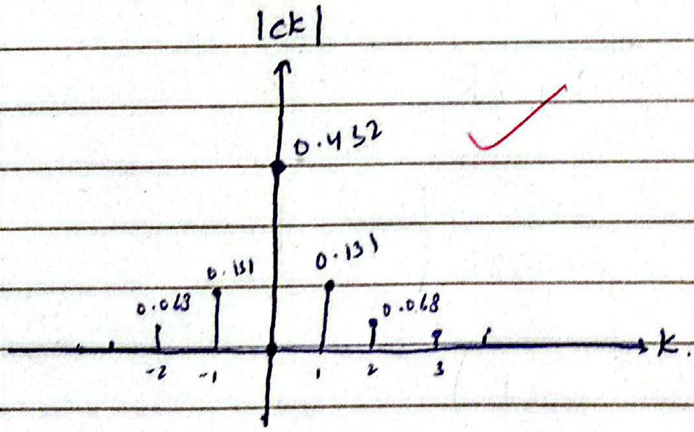
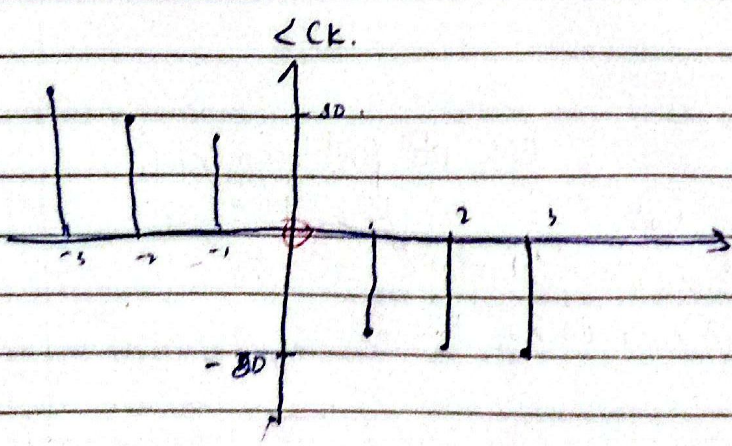
$$c_k = \frac{0.864}{2 + j2\pi k}$$

k	c_k	$\angle c_k$
0	0.432	0
1	0.131	-72.34
2	0.068	-80.96
3	0.045	-83.94
∞	0	-90 pure imaginary.

$$\lim_{k \to \infty} \left(\frac{0.864}{2 + j2\pi k} \right)$$





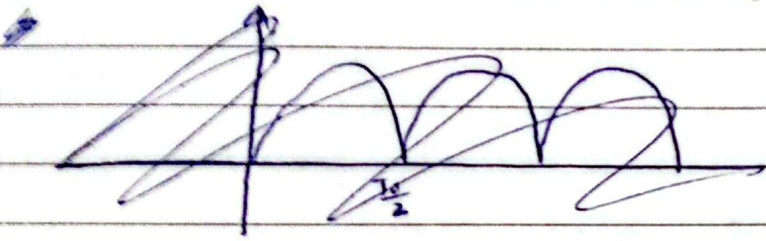
$k \to \infty$

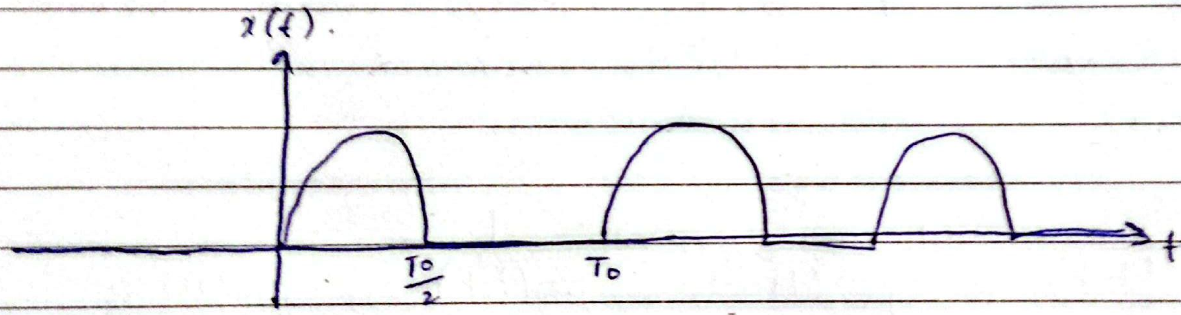
$\angle c_k = 90.$

Q.3)

$$\cancel{c_k = \frac{1}{4\pi(k-1)}\left[e^{-j\pi(k-1)} - 1\right] - \frac{1}{4\pi(k+1)}\left[e^{-j\pi(k+1)} - 1\right]}$$

$$\cancel{= \frac{e^{-j\pi(k-1)}}{4\pi(k-1)} - \frac{1}{4\pi(k-1)} - \frac{e^{-j\pi(k+1)}}{4\pi(k+1)} + \frac{1}{4\pi(k+1)}}$$





$$c_k = \frac{2}{T_0}\int_0^{\frac{T_0}{2}} \sin(w_0 t)\, e^{-jkw_0 t}\, .dt$$

$w_0 = \frac{2\pi}{T_0}\left(\frac{T_0}{2}\right)$

$$\cancel{\frac{2}{T_0}\left[\frac{e^{-jkw_0 t}}{(kw_0)^2 + (w_0)^2}\left[-jkw_0 \sin(w_0 t) - w_0 \cos\right]\right]}$$

$$= \frac{2}{T_0}\int_0^{\frac{T_0}{2}} \left(\frac{e^{jw_0 t} - e^{-jw_0 t}}{2j}\right) e^{-jkw_0 t}\, .dt$$

$$\Rightarrow \frac{1}{jT_0}\int_0^{\frac{T_0}{2}} e^{jw_0 t - jkw_0 t} - e^{-jw_0 t - jkw_0 t}\, .dt$$

$$\Rightarrow \frac{-j}{T_0}\left[\frac{e^{jw_0 t - jkw_0 t}}{jw_0 - jkw_0} + \frac{e^{-jw_0 t - jkw_0 t}}{jw_0 + jkw_0}\right]_0^{\frac{T_0}{2}}$$

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$$\Rightarrow \cancel{\frac{-j}{T_0}\left[\frac{e^{j\pi}e^{-j\pi k}}{j\omega_0(1-k)} + \frac{e^{-j\pi-j\pi k}}{j\omega_0(1+k)} - \frac{1}{j\omega_0(1-k)} - \frac{1}{j\omega_0(1+k)}\right]}$$

$$\Rightarrow \cancel{\frac{-1}{T_0}\left[\frac{e^{j\pi}e^{-j\pi k}}{\omega_0(1-k)} + \frac{e^{-j\pi-j\pi k}}{\omega_0(1+k)} - \frac{1}{\omega_0(1-k)} - \frac{1}{j\omega_0(1+k)}\right]}$$

$$\Rightarrow \cancel{\frac{-e^{j\pi}e^{-j\pi k}}{2\pi(1-k)} - \frac{e^{-j\pi-j\pi k}}{2\pi(1+k)} + \frac{1}{2\pi(1-k)} + \frac{1}{2\pi(1+k)}}$$

$\Rightarrow$

$$\Rightarrow \cancel{\frac{-1}{T_0}\left[\frac{e^{j\pi-j\pi k}}{j\pi-j\pi k} + \frac{e^{-j\pi-j\pi k}}{j\pi+j\pi k} - \frac{1}{j}\right]}$$

15/15

$$\Rightarrow \frac{-j}{T_0}\left[\frac{e^{j\pi-j\pi k}}{j\omega_0(1-k)} + \frac{e^{-j\pi-j\pi k}}{j\omega_0(1+k)} - \frac{1}{j\omega_0(1-k)} - \frac{1}{j\omega_0(1+k)}\right]$$

$$\Rightarrow \frac{+e^{j\pi-j\pi k}}{2\pi(k-1)} - \frac{e^{-j\pi-j\pi k}}{2\pi(1+k)} - \frac{1}{2\pi(k-1)} + \frac{1}{2\pi(1+k}$$

$$\Rightarrow \frac{e^{-j\pi(k-1)}}{2\pi(k-1)} + \frac{1}{2\pi(k-1)} - \left(\frac{e^{-j\pi(1+k)}}{2\pi(1+k)} - \frac{1}{2\pi(1+k)}\right)$$

$$\Rightarrow \frac{1}{2\pi(k-1)}\left[e^{-j\pi(k-1)} - 1\right] - \frac{1}{2\pi(k+1)}\left[e^{-j\pi(k+1)} - 1\right]$$

4

4